Explicit involution classes for the classical part of L045

Private proof component: publication_recovery

This note proves the lower estimates needed in Section 4 of the fixed L045 source, and corrects its special unitary degree-four estimate. It does not establish the required maximal-overgroup or automorphism bounds, or assert a complete recovery of the original paper. Throughout, $\mathrm{i}_2(H)$ counts elements of order exactly two. The finite 35 parameter certificates are a separate, already distinctly reviewed component.

Theorem 1. *For every prime power q, the following strict lower bounds hold:*

G	*range*	$\mathrm{i}_2(G) > q^D/4$
$\mathrm{PSL}_n(q),\ \mathrm{PSU}_n(q)$	$n \geq 4$	$D = \lfloor n^2/2 \rfloor$
$\mathrm{PSp}_{2m}(q)$	$m \geq 2$	$D = m^2 + m$
$\Omega_{2m+1}(q)$	$m \geq 3,\ q$ *odd*	$D = m^2 + m$
$\mathrm{P}\,\Omega^{\epsilon}_{2m}(q)$	$m \geq 4,\ \epsilon = \pm 1$	$D = m^2 - 1.$

The nonsimple group $\mathrm{Sp}_4(2)$ is allowed in this numerical statement; it is not a positive simple-group obligation in L045.

Classical conventions and elementary counts. We use the natural finite-field matrix groups. In odd characteristic, Ω is the kernel of the spinor norm in the determinant-one orthogonal group; in even characteristic it is the kernel of the Dickson invariant $g \mapsto \mathrm{rank}(g - I) \bmod 2$. These are the usual matrix descriptions of the groups denoted Ω in the source. A projective group means the quotient by the scalar centre. The ordinary finite-field form conventions, including the uniqueness of a nondegenerate hermitian form in each dimension and the two signs of an even-dimensional quadratic form, are retained elementary classical-group inputs. No maximal-subgroup or involution-class classification is used below.

Put $P_j(q) = \prod_{i=1}^{j}(1 - q^{-2i})$ and $F_j(q) = \prod_{i=1}^{j}(1 - (-q)^{-i})$, with empty products one. Counting ordered bases, symplectic bases, orthogonal bases, and unitary bases gives

$$|\,\mathrm{GL}_j(q)| = q^{j^2}\prod_{i=1}^{j}(1 - q^{-i}), \qquad |\,\mathrm{GU}_j(q)| = q^{j^2}F_j(q),$$

$$|\,\mathrm{Sp}_{2j}(q)| = q^{2j^2+j}P_j(q), \qquad |O^{\epsilon}_{2j}(q)| = 2q^{2j^2-j}(1 - \epsilon q^{-j})P_{j-1}(q),$$

$$|O_{2j+1}(q)| = 2q^{2j^2+j}P_j(q) \qquad (q \text{ odd}).$$

We count whole ambient conjugacy orbits of matrices lying in the desired special group. They need not be single conjugacy classes of that special group. If X^2 is scalar, at most two scalar multiples of X can lie in its ambient class: $(zX)^2 = X^2$ forces $z^2 = 1$. If the two eigenvalue multiplicities are unequal, only $z = 1$ is possible. Thus the divisions by two below are valid lower bounds even when special-group classes split.

Lemma 2. *For $q \geq 2$, $F_j(q) \leq 1 + q^{-1}$. A consecutive interval of its factors starting at a has product at least $1 - q^{-a}$, and has product greater than one if a is odd. Also*

$$\prod_{i \geq a}(1 - q^{-2i}) > 1 - \frac{q^{-2a}}{1 - q^{-2}}, \qquad \prod_{\substack{i \geq 1 \\ i\ odd}}(1 - q^{-i}) > 1 - \frac{q}{q^2 - 1}.$$

In the last inequality we use $q \geq 3$; the finite products used below obey the same strict estimates.

Proof. Pair an even factor $1 - q^{-2i}$ with its following odd factor to obtain a product less than one; this proves the upper bound for F_j. Pair an odd factor with its following even factor to obtain a product greater than one; an interval beginning at an even index loses only its first factor. The remaining inequalities follow by expanding a product of numbers $1 - a_i$, $0 < a_i < 1$, and bounding it below by $1 - \sum a_i$. □

Linear and unitary groups in odd characteristic. For odd n choose an even k nearest $n/2$ and take the matrix with -1-eigenspace of dimension k. For even $n = 2m$ with even m, take the balanced $m + m$ decomposition. All these matrices have determinant one; use orthogonal hermitian summands in the unitary case. Their ambient orbits have sizes $|\operatorname{GL}_n|/(|\operatorname{GL}_k||\operatorname{GL}_{n-k}|)$ or $|\operatorname{GU}_n|/(|\operatorname{GU}_k||\operatorname{GU}_{n-k}|)$, respectively. The normalized linear ratio exceeds one. For the unitary ratio, $k, n-k \geq 2$, so Lemma 2 gives

$$\frac{F_n(q)}{F_k(q)F_{n-k}(q)} > \frac{1-q^{-3}}{1+q^{-1}} \geq \frac{13}{18}.$$

The orbit exponent is $2k(n-k) = \lfloor n^2/2 \rfloor$. Division by at most two still leaves a constant exceeding $1/4$.

If $n = 2m$ and m is odd, use a determinant-one matrix X with $X^2 = -I$. In the linear case with $q \equiv 1 \pmod 4$, take $\operatorname{diag}(aI_m, -aI_m)$, $a^2 = -1$; with $q \equiv 3 \pmod 4$, take multiplication by a on $\mathbb{F}_{q^2}^m$. The centralizers are $\operatorname{GL}_m(q)^2$ and $\operatorname{GL}_m(q^2)$. In the unitary case the two congruences are reversed: a has norm one if $q \equiv 3 \pmod 4$, while for $q \equiv 1 \pmod 4$ the two eigenspaces are paired totally isotropic spaces, and the centralizer is $\operatorname{GL}_m(q^2)$. In the latter cases the respective normalized orbit ratios are

$$\prod_{\substack{1 \leq i \leq 2m-1 \\ i \text{ odd}}} (1-q^{-i}) > 5/8, \qquad \prod_{\substack{1 \leq i \leq 2m-1 \\ i \text{ odd}}} (1+q^{-i}) > 1.$$

The exponent is $2m^2$, and projectivization loses at most two.

Linear and unitary groups in characteristic two. Write $n = 2k+l$, $l \in \{0,1\}$. A linear involution of Jordan shape $(2^k, 1^l)$ has centralizer $q^{k(2n-3k)} \operatorname{GL}_k(q) \operatorname{GL}_l(q)$. This is obtained by writing commuting endomorphisms along its image/kernel filtration: the invertible diagonal blocks are one repeated k-block and one l-block. Its normalized orbit ratio is

$$\frac{\prod_{i=k+1}^{n}(1-q^{-i})}{\prod_{i=1}^{l}(1-q^{-i})} > 3/4.$$

For a hermitian space, put $X = I + N$. Unitarity and $N^2 = 0$ imply that $W = \operatorname{im} N$ is totally isotropic and N is self-adjoint. Conversely, for each isotropic k-space W, the rank-k self-adjoint maps $V/W^\perp \longrightarrow W$ are precisely the nonsingular hermitian matrices; their number is $|\operatorname{GL}_k(q^2)|/|\operatorname{GU}_k(q)|$. The stabilizer of W has order $q^{k(2n-3k)}|\operatorname{GL}_k(q^2)||\operatorname{GU}_l(q)|$: the radical has $2kl + k^2$ parameters over $\mathbb{F}_q$, and the indicated two diagonal blocks are free. Therefore the total number is

$$\frac{|\operatorname{GU}_n(q)|}{q^{k(2n-3k)}|\operatorname{GU}_k(q)||\operatorname{GU}_l(q)|}.$$

Its exponent is $2k(n-k)$ and its normalized ratio $F_n/(F_kF_l)$ exceeds $(1-2^{-3})/(1+2^{-1}) = 7/12$. All matrices just counted are unipotent, hence determinant one. In characteristic two no nontrivial scalar multiple of a unipotent matrix is unipotent. This proves both projective lower bounds.

Symplectic groups in odd characteristic. Use a symplectic complex structure $X^2 = -I$. If $q \equiv 1 \pmod 4$, its two isotropic eigenspaces are paired and its centralizer is $\operatorname{GL}_m(q)$. If $q \equiv 3 \pmod 4$, view the space as a hermitian m-space over $\mathbb{F}_{q^2}$ and obtain centralizer $\operatorname{GU}_m(q)$. These statements follow directly by requiring a commuting map to preserve the pairing or the hermitian form, respectively. The orbit exponent is $m^2 + m$; the normalized ratios are

$$\prod_{i=1}^{m}(1+q^{-i}) > 1, \qquad \prod_{i=1}^{m}(1+(-q)^{-i}) > 5/8.$$

At most two-to-one scalar fusion gives the result.

Symplectic groups in characteristic two. Fix an isotropic k-space W. The symplectic rank-k square-zero maps with image W correspond to nonsingular symmetric $k \times k$ matrices. Retain just the nonalternating ones. Every nonalternating nondegenerate symmetric form over the perfect field $\mathbb{F}_q$ is equivalent to I_k. One may diagonalize after splitting a nonsingular vector; a remaining alternating plane together with one nonsingular line is equivalent to three nonsingular lines. Its isometry group has order

$$J_k(q) = \begin{cases} |\operatorname{Sp}_{k-1}(q)|, & k \text{ odd}, \\ q^{k-1}|\operatorname{Sp}_{k-2}(q)|, & k \text{ even}. \end{cases}$$

Indeed the distinguished vector representing $v \mapsto \sqrt{(v,v)}$ is fixed. If it is nonsingular its perpendicular complement is symplectic; if it is singular the quotient of its perpendicular space is symplectic and the kernel, by the two remaining row/column equations, has q^{k-1} elements. The isotropic-space stabilizer has Levi part $\mathrm{GL}_k(q)\,\mathrm{Sp}_{2m-2k}(q)$ and radical order q^u, where $u = (4mk - 3k^2 + k)/2$ (count the free off-diagonal block and its symmetric correction). Consequently the number of these involutions is exactly

$$C_{m,k}(q) = \frac{|\,\mathrm{Sp}_{2m}(q)|}{q^u|\,\mathrm{Sp}_{2m-2k}(q)|J_k(q)}. \tag{1}$$

For $k = m$ this is q^{m^2+m} times $\prod_{i=\lceil m/2\rceil}^{m}(1 - q^{-2i}) > 2/3$. The symplectic centre is trivial in characteristic two.

Orthogonal groups in odd characteristic. For odd dimension $2m + 1$, let $2a$ be the even member of $\{m, m + 1\}$, and $b = m - a$. Choose a nondegenerate $2a$-space with square discriminant; such a space of one of the two signs exists with a nondegenerate complement. Acting as $-I$ on it and I on its complement has determinant and spinor norm one. The full orthogonal orbit has size

$$\frac{|O_{2m+1}(q)|}{|O^{\delta}_{2a}(q)||O_{2b+1}(q)|}.$$

All its elements lie in Ω. The exponent is $m^2 + m$, and its normalized ratio exceeds $(1/2)(7/8)/(4/3) = 21/64$.

For dimension $2m$, put $a = \lfloor m/2\rfloor$, $b = m - a$, choose again a square-discriminant $2a$-space, and use the same sign involution. Its orthogonal orbit has size

$$\frac{|O^{\epsilon}_{2m}(q)|}{|O^{\delta}_{2a}(q)||O^{\epsilon\delta}_{2b}(q)|}.$$

The exponent is $4ab$. Apart from the leading factor $1/2$, its normalized ratio is greater than

$$\frac{(1 - 3^{-4})(7/8)}{(1 + 3^{-2})^2} = 7/10.$$

If m is odd the two multiplicities differ, so the projective map is injective on this orbit and $4ab = m^2 - 1$. If m is even it loses at most two and $4ab = m^2$; the extra factor $q \geq 3$ more than pays this loss.

Orthogonal groups in characteristic two. Use (1) with even $k = m$ for even m, and $k = m - 1$ for odd m. In a hyperbolic symplectic basis the chosen involution sends f_i to $f_i + e_i$ for $1 \leq i \leq k$ and fixes all other basis vectors. A quadratic form Q with this polar form is invariant precisely when $Q(e_i) = 1$ for these k indices. Thus exactly q^{2m-k} forms are invariant. Their Arf trace is $\mathrm{Tr}(\sum Q(e_i)Q(f_i))$; since $k > 0$, one free $Q(f_i)$ makes its two values equally frequent. Exactly $q^{2m-k}/2$ invariant forms have each sign. There are $q^m(q^m + \epsilon)/2$ forms of sign ϵ with the fixed polar form. Double counting pairs (X, Q) gives

$$C_{m,k}(q)\frac{q^{m-k}}{q^m + \epsilon} \tag{2}$$

involutions in each orthogonal group. Their Dickson invariant is $k \bmod 2 = 0$, so all belong to Ω; its scalar centre is trivial here. For even m, (2) is

$$q^{m^2}\frac{\prod_{i=m/2}^{m}(1 - q^{-2i})}{1 + \epsilon q^{-m}}.$$

For odd m it is

$$q^{m^2-1}\frac{\prod_{i=(m-1)/2}^{m}(1 - q^{-2i})}{(1 - q^{-2})(1 + \epsilon q^{-m})}.$$

Both normalized ratios exceed $(11/12)/(17/16) = 44/51$. This finishes Theorem 1.

Proposition 3. *If $q \equiv 3 \pmod 4$, then*

$$\mathrm{i}_2(\mathrm{PSU}_4(q)) = \frac{q^4(q^2 - q + 1)(q^2 + 1)}{2} + \begin{cases} 0, & q \equiv 3 \pmod 8, \\ q^3(q - 1)(q^2 + 1), & q \equiv 7 \pmod 8. \end{cases}$$

In particular the source's special assertion $\mathrm{i}_2(\mathrm{PSU}_4(q)) > q^8/2$ is false for all such q.

Proof. A projective involution lifts to $X \in \mathrm{SU}_4(q)$ with $X^2 = zI$, $z \in \mu_4$. It is semisimple. For $z = 1, -1$, determinant one forces the noncentral multiplicities to be $2+2$. Both choices give the same projective family; its size is $|\,\mathrm{GU}_4\,|/(2|\,\mathrm{GU}_2\,|^2)$, the first summand displayed above. For z of order four, choose $a^2 = z$, so a has order eight. If $q \equiv 3 \pmod 8$, $a^{q+1} = -1$; its two eigenspaces must be paired isotropic spaces of dimensions $2+2$. But the determinant is then $a^4 = -1$, impossible. If $q \equiv 7 \pmod 8$, $a^{q+1} = 1$; the eigenspaces are nondegenerate and determinant one requires multiplicities $1+3$ or $3+1$. Scalar multiplication cycles all four choices of a and identifies the two multiplicity orientations, leaving one projective family. Within one $1+3$ ambient orbit the projective map is injective, so its size is $|\,\mathrm{GU}_4\,|/(|\,\mathrm{GU}_1\,||\,\mathrm{GU}_3\,|) = q^3(q-1)(q^2+1)$. There are no remaining possibilities for z or for the eigenspaces.

For $q \equiv 3 \pmod 8$, subtracting $q^8/2$ from the first summand gives $-(q^4/2)(q(q-1)^2-1) < 0$. For $q \equiv 7 \pmod 8$, the difference is $-(q^3/2)(q^4-4q^3+3q^2-3q+2) < 0$, since $q \geq 7$. □

Exact controls and attribution. The accompanying standard-library program evaluates all written class formulas with rational arithmetic at $1,619$ parameter choices, verifies the strict quarter bounds, and checks the correction at seven fields. These finite checks are controls on the formulas, not premises for the uniform inequalities. The ordinary character table $\mathrm{U}_4(3)$ gives $2,835$ involutions, agreeing with Proposition 3; at $q = 7$ the formula gives $2,683,975 < 7^8/2$. A fresh clean table replay is retained with the component. The construction and calculations in this note are new recovery work by publication_recovery, responding to root's request for explicit lower estimates. The fixed source is Leemans–Liebeck, *Chiral polyhedra and finite simple groups*, arXiv:1603.07713v3, Lemma 2.4 and Section 4 (especially printed page 9). No claim of novelty or priority is made for the classical formulas. The remaining existence argument must separately pay every bad-generator and reverser bound.